1.1 Propositional Logic

Exercises 2,16, and 34(a, d)

SECTION 1.1 Propositional Logic

2. Propositions must have clearly defined truth values, so a proposition must be a declarative sentence with no free variables.

a) This is not a proposition; it's a command.

b) This is not a proposition; it's a question.

c) This is a proposition that is false, as anyone who has been to Maine knows.

d) This is not a proposition; its truth value depends on the value of x.

e) This is a proposition that is false.

f) This is not a proposition; its truth value depends on the value of n.

16. Let p, q, and r be the propositions

p: You get an A on the final exam.
q: You do every exercise in this book.
r: You get an A in this class.

Write these propositions using p, q, and r and logical connectives (including negations).

a) You get an A in this class, but you do not do every exercise in this book.

b) You get an A on the final, you do every exercise in this book, and you get an A in this class.

c) To get an A in this class, it is necessary for you to get an A on the final.

d) You get an A on the final, but you don't do every exercise in this book; nevertheless, you get an A in this class.

e) Getting an A on the final and doing every exercise in this book is sufficient for getting an A in this class.

f) You will get an A in this class if and only if you either do every exercise in this book or you get an A on the final.

a) $r \wedge \neg q$ **b)** $p \wedge q \wedge r$ **c)** $r \rightarrow p$ **d)** $p \wedge \neg q \wedge r$ **e)** $(p \wedge q) \rightarrow r$ **f)** $r \leftrightarrow (q \vee p)$

34. Construct a truth table for each of these compound propositions.

a) $p \rightarrow \neg p$ **b)** $p \leftrightarrow \neg p$
c) $p \oplus (p \vee q)$ **d)** $(p \wedge q) \rightarrow (p \vee q)$
e) $(q \rightarrow \neg p) \leftrightarrow (p \leftrightarrow q)$
f) $(p \leftrightarrow q) \oplus (p \leftrightarrow \neg q)$

To construct the truth table for a compound proposition, we work from the inside out. In each case, we will show the intermediate steps. In part **(d)**, for example, we first construct the truth tables for $p \wedge q$ and for $p \vee q$ and combine them to get the truth table for $(p \wedge q) \rightarrow (p \vee q)$. For parts **(a)** and **(b)** we have the following table (column three for part **(a)**, column four for part **(b)**).

p	$\neg p$	$p \rightarrow \neg p$	$p \leftrightarrow \neg p$
T	F	F	F
F	T	T	F

For parts **(c)** and **(d)** we have the following table.

p	q	$p \vee q$	$p \wedge q$	$p \oplus (p \vee q)$	$(p \wedge q) \rightarrow (p \vee q)$
T	T	T	T	F	T
T	F	T	F	F	T
F	T	T	F	T	T
F	F	F	F	F	T

1.2 Applications of Propositional Logic

Exercises 2, 4, 7 and 12

SECTION 1.2 Applications of Propositional Logic

2. Recall that p only if q means $p \to q$. In this case, if you can see the movie then you must have fulfilled one of the two requirements. Therefore the statement is $m \to (e \vee p)$. Notice that in everyday life one might actually say "You can see the movie *if* you meet one of these conditions," but logically that is not what the rules really say.

4. The condition stated here is that if you use the network, then either you pay the fee or you are a subscriber. Therefore the proposition in symbols is $w \to (d \vee s)$.

7. a) $q \to p$ **b)** $q \wedge \neg p$ **c)** $q \to p$ **d)** $\neg q \to \neg p$

12. Are these system specifications consistent? "If the file system is not locked, then new messages will be queued. If the file system is not locked, then the system is functioning normally, and conversely. If new messages are not queued, then they will be sent to the message buffer. If the file system is not locked, then new messages will be sent to the message buffer. New messages will not be sent to the message buffer."

12. This system is consistent. We use L, Q, N, and B to stand for the basic propositions here, "The file system is locked," "New messages will be queued," "The system is functioning normally," and "New messages will be sent to the message buffer," respectively. Then the given specifications are $\neg L \to Q$, $\neg L \leftrightarrow N$, $\neg Q \to B$, $\neg L \to B$, and $\neg B$. If we want consistency, then we had better have B false in order that $\neg B$ be true. This requires that both L and Q be true, by the two conditional statements that have B as their consequence. The first conditional statement therefore is of the form $\mathrm{F} \to \mathrm{T}$, which is true. Finally, the biconditional $\neg L \leftrightarrow N$ can be satisfied by taking N to be false. Thus this set of specifications is consistent. Note that there is just this one satisfying truth assignment.

1.3 Propositional Equivalences

Exercises 2,6,11,21,30,66(a)

SECTION 1.3 Propositional Equivalences

2. There are two cases. If p is true, then $\neg(\neg p)$ is the negation of a false proposition, hence true. Similarly, if p is false, then $\neg(\neg p)$ is also false. Therefore the two propositions are logically equivalent.

6. We see that the fourth and seventh columns are identical.

p	q	$p \wedge q$	$\neg(p \wedge q)$	$\neg p$	$\neg q$	$\neg p \vee \neg q$
T	T	T	F	F	F	F
T	F	F	T	F	T	T
F	T	F	T	T	F	T
F	F	F	T	T	T	T

11. a)

p	q	$p \wedge q$	$(p \wedge q) \rightarrow p$
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	T

b)

p	q	$p \vee q$	$p \rightarrow (p \vee q)$
T	T	T	T
T	F	T	T
F	T	T	T
F	F	F	T

c)

p	q	$\neg p$	$p \rightarrow q$	$\neg p \rightarrow (p \rightarrow q)$
T	T	F	T	T
T	F	F	F	T
F	T	T	T	T
F	F	T	T	T

d)

p	q	$p \wedge q$	$p \rightarrow q$	$(p \wedge q) \rightarrow (p \rightarrow q)$
T	T	T	T	T
T	F	F	F	T
F	T	F	T	T
F	F	F	T	T

e)

p	q	$p \rightarrow q$	$\neg(p \rightarrow q)$	$\neg(p \rightarrow q) \rightarrow p$
T	T	T	F	T
T	F	F	T	T
F	T	T	F	T
F	F	T	F	T

f)

p	q	$p \rightarrow q$	$\neg(p \rightarrow q)$	$\neg q$	$\neg(p \rightarrow q) \rightarrow \neg q$
T	T	T	F	F	T
T	F	F	T	T	T
F	T	T	F	F	T
F	F	T	F	T	T

21. Show that $\neg(p \leftrightarrow q)$ and $p \leftrightarrow \neg q$ are logically equivalent.

The proposition ¬(p↔q) is true when p↔q is false, which means that p and q have different truth values. Because this is precisely when ¬p↔q is true, the two expressions are logically equivalent.

30. Show that $\neg p \rightarrow (q \rightarrow r)$ and $q \rightarrow (p \vee r)$ are logically equivalent.

Applying the third and first equivalences in Table 7, we have $\neg p \rightarrow (q \rightarrow r) \equiv p \vee (q \rightarrow r) \equiv p \vee \neg q \vee r$. Applying the first equivalence in Table 7 to $q \rightarrow (p \vee r)$ shows that $\neg q \vee p \vee r$ is equivalent to it. But these are equivalent by the commutative and associative laws.

TABLE 7 Logical Equivalences Involving Conditional Statements.

$p \rightarrow q \equiv \neg p \vee q$
$p \rightarrow q \equiv \neg q \rightarrow \neg p$
$p \vee q \equiv \neg p \rightarrow q$
$p \wedge q \equiv \neg(p \rightarrow \neg q)$
$\neg(p \rightarrow q) \equiv p \wedge \neg q$
$(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$
$(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$
$(p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$
$(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$

66. Determine whether each of these compound propositions is satisfiable.

a) $(p \vee q \vee \neg r) \wedge (p \vee \neg q \vee \neg s) \wedge (p \vee \neg r \vee \neg s) \wedge (\neg p \vee \neg q \vee \neg s) \wedge (p \vee q \vee \neg s)$

a) Since p occurs in four of the five disjunctions, we can make p true, and then make q false (and make r and s anything we please). Thus this proposition is satisfiable.

1.4 Predicates and Quantifiers

- 8
- 17
- 20(a, c, and d)
- 23
- 36
- 40(b, d, and e)
- 57

8. Note that part **(b)** and part **(c)** are not the sorts of things one would normally say.

a) If an animal is a rabbit, then that animal hops. (Alternatively, every rabbit hops.)

b) Every animal is a rabbit and hops.

c) There exists an animal such that if it is a rabbit, then it hops. (Note that this is trivially true, satisfied, for example, by lions, so it is not the sort of thing one would say.)

d) There exists an animal that is a rabbit and hops. (Alternatively, some rabbits hop. Alternatively, some hopping animals are rabbits.)

17.a) $P(0) \vee P(1) \vee P(2) \vee P(3) \vee P(4)$

b) $P(0) \wedge P(1) \wedge P(2) \wedge P(3) \wedge P(4)$

c) $\neg P(0) \vee \neg P(1) \vee \neg P(2) \vee \neg P(3) \vee \neg P(4)$

d) $\neg P(0) \wedge \neg P(1) \wedge \neg P(2) \wedge \neg P(3) \wedge \neg P(4)$

e) $\neg(P(0) \vee P(1) \vee P(2) \vee P(3) \vee P(4))$

f) $\neg(P(0) \wedge P(1) \wedge P(2) \wedge P(3) \wedge P(4))$

20.

a) We want to assert that $P(x)$ is true for some x in the domain, so either $P(-5)$ is true or $P(-3)$ is true or $P(-1)$ is true or $P(1)$ is true or $P(3)$ is true or $P(5)$ is true. Thus the answer is $P(-5) \vee P(-3) \vee P(-1) \vee P(1) \vee P(3) \vee P(5)$.

b) $P(-5) \wedge P(-3) \wedge P(-1) \wedge P(1) \wedge P(3) \wedge P(5)$

c) The formal translation is as follows: $((-5 \neq 1) \to P(-5)) \wedge ((-3 \neq 1) \to P(-3)) \wedge ((-1 \neq 1) \to P(-1)) \wedge ((1 \neq 1) \to P(1)) \wedge ((3 \neq 1) \to P(3)) \wedge ((5 \neq 1) \to P(5))$. However, since the hypothesis $x \neq 1$ is false when x is 1 and true when x is anything other than 1, we have more simply $P(-5) \wedge P(-3) \wedge P(-1) \wedge P(3) \wedge P(5)$.

d) The formal translation is as follows: $((-5 \geq 0) \wedge P(-5)) \vee ((-3 \geq 0) \wedge P(-3)) \vee ((-1 \geq 0) \wedge P(-1)) \vee ((1 \geq 0) \wedge P(1)) \vee ((3 \geq 0) \wedge P(3)) \vee ((5 \geq 0) \wedge P(5))$. Since only three of the x's in the domain meet the condition, the answer is equivalent to $P(1) \vee P(3) \vee P(5)$.

e) For the second part we again restrict the domain: $(\neg P(-5) \vee \neg P(-3) \vee \neg P(-1) \vee \neg P(1) \vee \neg P(3) \vee \neg P(5)) \wedge (P(-1) \wedge P(-3) \wedge P(-5))$. This is equivalent to $(\neg P(1) \vee \neg P(3) \vee \neg P(5)) \wedge (P(-1) \wedge P(-3) \wedge P(-5))$.

23.Let C(x) be the propositional function “x is in your class.”

a)$\exists x H(x)$and$\exists x(C(x) \wedge H(x))$,where H(x) is “x can speak Hindi”

b)$\forall x F(x)$ and $\forall x(C(x) \rightarrow F(x))$,where F(x) is “x is friendly”

c)$\exists x \neg B(x)$and$\exists x(C(x) \wedge \neg B(x))$,where B(x) is“x was born in California”

d)$\exists x M(x)$and$\exists x(C(x) \wedge M(x))$,where M(x) is“x has been in a movie”

e)$\forall x \neg L(x)$and$\forall x(C(x) \rightarrow \neg L(x))$,where L(x) is“x has taken a course in logic programming”

36.a) $\exists x((x \geqslant 3) \vee (x \leqslant -2))$

b) $\exists x((x \geqslant 5) \vee (x<0))$

c) $\forall x((x>1) \vee (x < -4))$

d) $\forall x((x \geqslant -1) \vee (x \leqslant -5))$

40. b) Every system is either malfunctioning or in a diagnostic state.

d) Some system is unavailable.

e) No system is working.

(We could also say “Every system is not working,” as long as we understood that this is different from “Not every system is working.”)

57. a) Yes

b) No

c)juana, kiko

d) math273, cs301

e) juana, kiko

1.5 Nested Quantifiers

- 3(a, c, and e)
- 9(a, c, g, and i)
- 12(c, h, and m)
- 14(b, d, e and f)
- 24(b and c)
- 26(d, e, and i)
- 35
- 39(b)
- 40(b)

3. a) There is some student in your class who has sent a message to some student in your class.

c) Every student in your class has sent a message to at least one student in your class.

e) Every student in your class has been sent a message from at least one student in your class.

9. a) ∀xL(x,Jerry)

c) ∃y∀xL(x, y)

g) ∃x(∀yL(y, x) ∧ ∀z((∀wL(w, z))→z = x))

i) ∀xL(x, x)

12.c) ¬C(Jan,Sharon)

h) ∃x∀y(x=y↔I(y))

m) ∃x∀yC(x,y)

14.

a) Let $S(x, y)$ mean that person x can speak language y. Then our statement is $\exists x\, S(x, \text{Hindi})$.

b) Let $P(x, y)$ mean that person x plays sport y. Then our statement is $\forall x \exists y\, P(x, y)$.

c) Let $V(x, y)$ mean that person x has visited state y. Then our statement is $\exists x(V(x, \text{Alaska}) \wedge \neg V(x, \text{Hawaii}))$.

d) Let $L(x, y)$ mean that person x has learned programming language y. Then our statement is $\forall x \exists y\, L(x, y)$.

e) Let $T(x, y)$ mean that person x has taken course y, and let $O(y, z)$ mean that course y is offered by department z. Then our statement is $\exists x \exists z \forall y(O(y, z) \rightarrow T(x, y))$.

f) Let $G(x, y)$ mean that persons x and y grew up in the same town. Then our statement is $\exists x \exists y(x \neq y \wedge G(x, y) \wedge \forall z(G(x, z) \rightarrow (x = y \vee x = z)))$.

g) Let $C(x, y, z)$ mean that persons x and y have chatted with each other in chat group z. Then our statement is $\forall x \exists y \exists z(x \neq y \wedge C(x, y, z))$.

24.

b) A nonnegative number minus a negative number is positive.

c) The difference of two nonpositive numbers is not necessarily nonpositive.

26. **a)** This is false, since $1 + 1 \neq 1 - 1$. **b)** This is true, since $2 + 0 = 2 - 0$.

c) This is false, since there are many values of y for which $1 + y \neq 1 - y$.

d) This is false, since the equation $x + 2 = x - 2$ has no solution.

e) This is true, since we can take $x = y = 0$. **f)** This is true, since we can take $y = 0$ for each x.

g) This is true, since we can take $y = 0$. **h)** This is false, since part **(d)** was false.

i) This is certainly false.

35.

Any domain with four or more members makes the statement true;

any domain with three or fewer members makes the statement false.

39. b) x = −4

40.

b) We can rewrite $y^2 - x < 100$ as $y^2 < 100 + x$. Since squares can never be negative, no such y exists if x is, say, -200. This x provides a counterexample.

1.6 Rules of Inference

- 3
- 8
- 20
- 27

3. a) Addition
b) Simplification
c) Modus ponens
d) Modus tollens
e) Hypothetical syllo gism

8. First we use universal instantiation to conclude from "For all x, if x is a man, then x is not an island" the special case of interest, "If Manhattan is a man, then Manhattan is not an island." Then we form the contrapositive (using also double negative): "If Manhattan is an island, then Manhattan is not a man." Finally we use modus ponens to conclude that Manhattan is not a man. Alternatively, we could apply modus tollens.

20. **a)** This is invalid. It is the fallacy of affirming the conclusion. Letting $a = -2$ provides a counterexample.
b) This is valid; it is modus ponens.

27.

Step	Reason
1. $\forall x(P(x) \wedge R(x))$	Premise
2. $P(a) \wedge R(a)$	Universal instantiation from (1)
3. $P(a)$	Simplification from (2)
4. $\forall x(P(x) \rightarrow (Q(x) \wedge S(x)))$	Premise
5. $Q(a) \wedge S(a)$	Universal modus ponens from(3) and (4)
6. $S(a)$	Simplification from (5)
7. $R(a)$	Simplification from (2)
8. $R(a) \wedge S(a)$	Conjunction from (7) and (6)
9. $\forall x(R(x) \wedge S(x))$	Universal generalization from (5)

1.7 Introduction to Proofs

- 3
- 17
- 37

3.
Suppose that n is even. Then n = 2k for some integer k. Therefore, $n^2=(2k)^2=4k^2 =2(2k^2)$. Because we have written n^2 as 2 times an integer, we conclude that n^2 is even.

17.
Assume that it is not true that $x \geqslant 1$ or $y \geqslant 1$. Then $x < 1$ and $y < 1$. Adding these two inequalities, we obtain $x + y < 2$, which is the negation of $x+y \geqslant 2$.

37.
No.

1.8 Proof Methods and Strategy

- 1
- 20
- 33

1.

$1^2 + 1 = 2 \geqslant 2=2^1$;

$2^2 + 1 = 5 \geqslant 4=2^2$;

$3^2 + 1 = 10 \geqslant 8 = 2^3$;

$4^2 +1 = 17 \geqslant 16 = 2^4$

20.

Given r, let a be the closest integer to r less than r, and let b be the closest integer to r greater than r. In the notation to be introduced in Section 2.3, $a = \lfloor r \rfloor$ and $b = \lceil r \rceil$. In fact, $b = a + 1$. Clearly the distance between r and any integer other than a or b is greater than 1 so cannot be less than $1/2$. Furthermore, since r is irrational, it cannot be exactly half-way between a and b, so exactly one of $r - a < 1/2$ and $b - r < 1/2$ holds.

33.

Because $5^4 = 625$, both x and y must be less than 5.Then $x^4 + y^4 \leqslant 4^4+4^4=512<625$.

2.1 Sets

- 22
- 23
- 25
- 47

22.a) 1

b) 1

c) 2

d)3

23. a){∅, {a}}

b){∅, {a}, {b}, {a, b}}

c) {∅,{∅},{{∅}},{∅,{∅}}}

25. a) 8

b) 16

c) 2

47. a) {−1,0,1}

b) Z−{0,1}

c) ∅

2.2 Set Operations

- 20 (a, and c)
- 27
- 63

20.

a) Suppose that $x \in A \cup B$. Then either $x \in A$ or $x \in B$. In either case, certainly $x \in A \cup B \cup C$. This establishes the desired inclusion.

c) Suppose that $x \in (A - B) - C$. Then x is in $A - B$ but not in C. Since $x \in A - B$, we know that $x \in A$ (we also know that $x \notin B$, but that won't be used here). Since we have established that $x \in A$ but $x \notin C$, we have proved that $x \in A - C$.

27. a) {4,6}

b) {0,1,2,3,4,5,6,7,8,9,10}

c) {4,5,6,8,10}

d) {0,2,4,5,6,7,8,9,10}

63.a)11111000000000000000000000 ∨ 01110010000000010001010000=111110100000000 10001010000,representing{a,b,c,d,e,g,p,t,v}

b)11111000000000000000000000 ∧ 01110010000000010001010000=011100000000000000 00000000,representing{b,c,d}

c)(11111000000000000000000000 ∨ 00011001100001100001100110) ∧ (0111001000000001 0001010000 ∨ 00101000100000100000100111)=11111001100001100001100110 ∧ 0111101 0100000110001110111=01111000100000100001100110,representing{b,c,d,e,i,o,t,u,x,y}

d)11111000000000000000000000 ∨ 01110010000000010001010000 ∨ 00101000100000100 000100111 ∨ 00011001100001100001100110=11111011100001110001110111,representing {a,b,c,d,e,g,h,i,n,o,p,t,u,v,x,y,z}

2.3 Functions

- 8
- 23
- 69 (a and f)

8. We simply round up or down in each case.

a) 1 **b)** 2 **c)** -1 **d)** 0 **e)** 3 **f)** -2 **g)** $\lfloor \frac{1}{2} + 1 \rfloor = \lfloor \frac{3}{2} \rfloor = 1$

h) $\lceil 0 + 1 + \frac{1}{2} \rceil = \lceil \frac{3}{2} \rceil = 2$

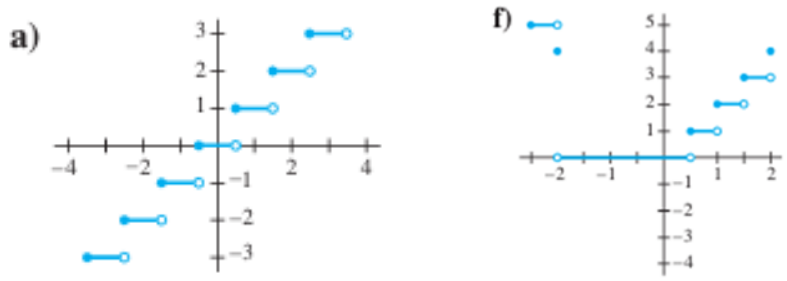
23. a) Yes

b) No

c) Yes

d) No

69.



2.4 Sequences and Summations

- 3
- 29

3.a) $a_0 = 2, a_1 = 3, a_2=5, a_3=9$

b) $a_0= 1, a_1=4, a_2=27, a_3 = 256$

c) $a_0 =0, a_1 =0, a_2 =1, a_3 =1$

d) $a_0 =0, a_1 =1, a_2 = 2, a_3 = 3$

29. a) 20

b)11

c)30

d)511

2.5 Cardinality of Sets

- 1
- 10

1. a) Countably infinite, $-1, -2, -3, -4, \cdots$

b) Countably infinite, $0, 2, -2, 4, -4, \cdots$

c)Countably infinite, $99, 98, 97, \cdots$

d) Uncountable

e) Finite

f) Countably infinite, $0, 7, -7, 14, -14, \cdots$ 3

10. In each case, let us take A to be the set of real numbers.

a) We can let B be the set of real numbers as well; then $A - B = \emptyset$, which is finite.

b) We can let B be the set of real numbers that are not positive integers; in symbols, $B = A - \mathbf{Z}^+$. Then $A - B = \mathbf{Z}^+$, which is countably infinite.

c) We can let B be the set of positive real numbers. Then $A - B$ is the set of negative real numbers and 0, which is certainly uncountable.